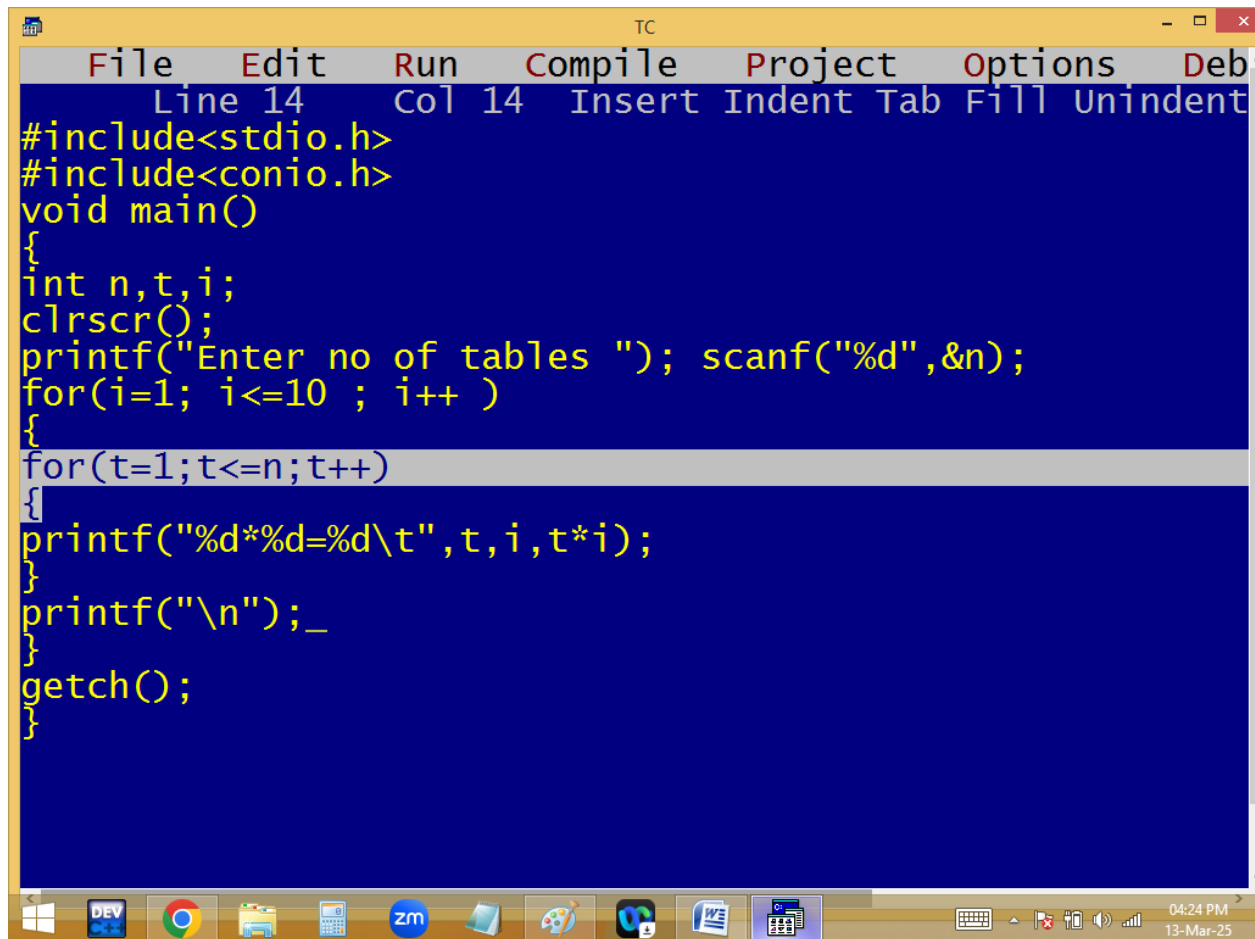


Tables side by side:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 14", "Col 14", and a list of actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. The code is as follows:

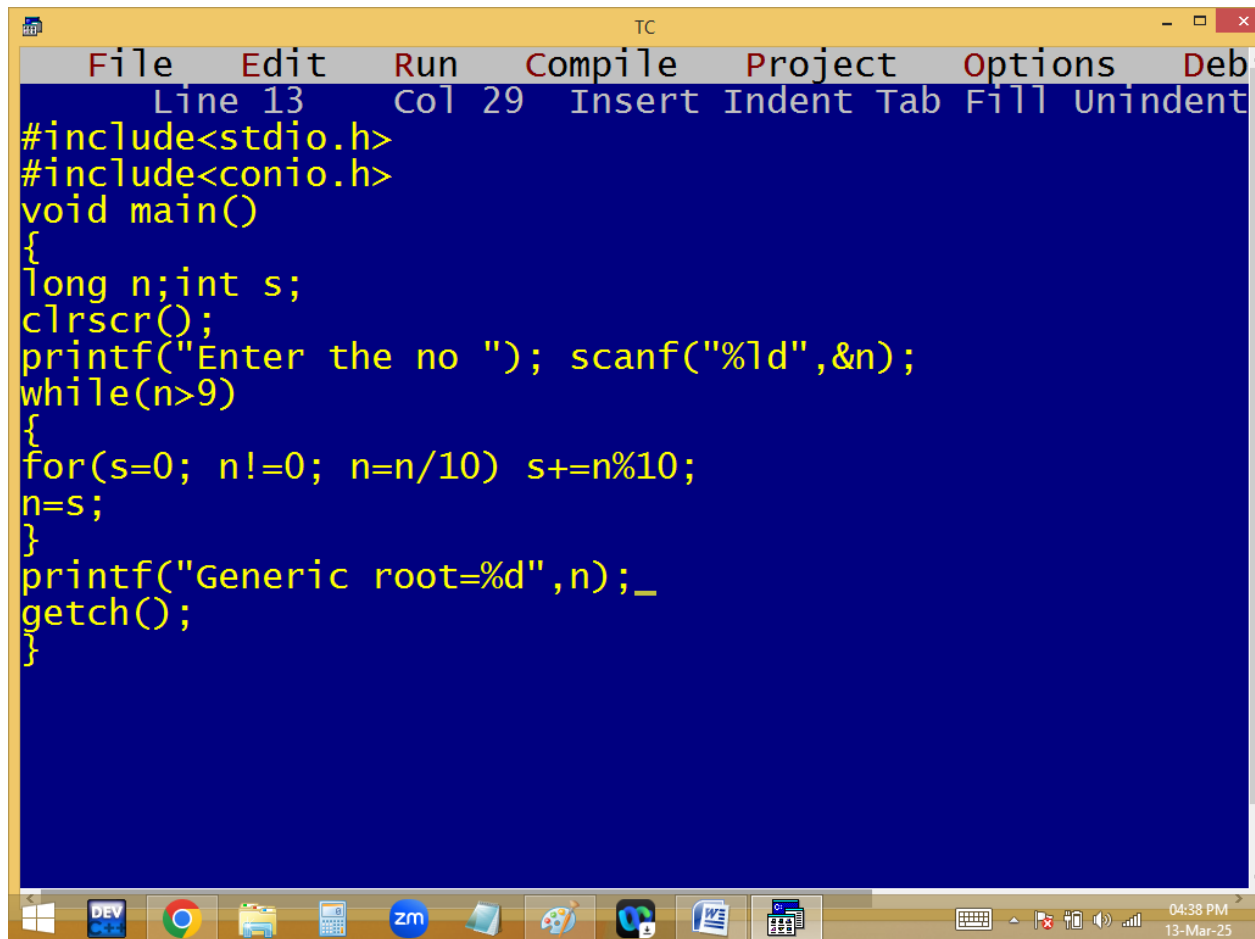
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,t,i;
    clrscr();
    printf("Enter no of tables "); scanf("%d",&n);
    for(i=1; i<=10 ; i++ )
    {
        for(t=1;t<=n;t++)
        {
            printf("%d*%d=%d\t",t,i,t*i);
        }
        printf("\n");_
    }
    getch();
}
```

The taskbar at the bottom of the screen contains several icons: Windows Start button, DEV icon, Google Chrome, File Explorer, Calculator, Zoom, a folder icon, a game controller icon, a VS Code icon, a Word document icon, and a calculator icon. On the right side of the taskbar, the system clock shows "04:24 PM" and the date "13-Mar-25".

```
TC
Enter no of tables 5
1*1=1 2*1=2 3*1=3 4*1=4 5*1=5
1*2=2 2*2=4 3*2=6 4*2=8 5*2=10
1*3=3 2*3=6 3*3=9 4*3=12 5*3=15
1*4=4 2*4=8 3*4=12 4*4=16 5*4=20
1*5=5 2*5=10 3*5=15 4*5=20 5*5=25
1*6=6 2*6=12 3*6=18 4*6=24 5*6=30
1*7=7 2*7=14 3*7=21 4*7=28 5*7=35
1*8=8 2*8=16 3*8=24 4*8=32 5*8=40
1*9=9 2*9=18 3*9=27 4*9=36 5*9=45
1*10=10 2*10=20 3*10=30 4*10=40 5*10=50
```

Find the generic root of given no:

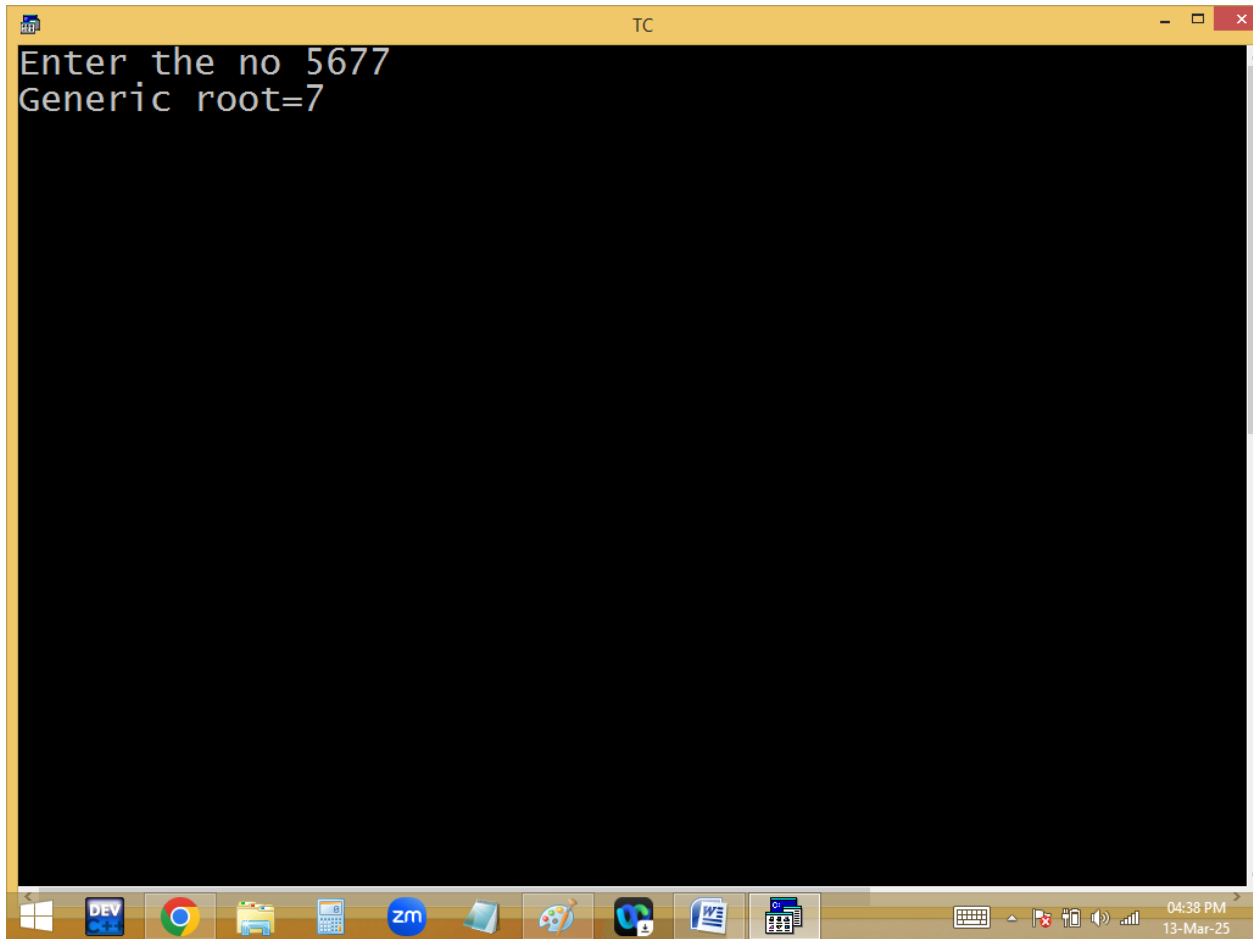
$$5+6+7+7=25 \rightarrow 2+5=7$$

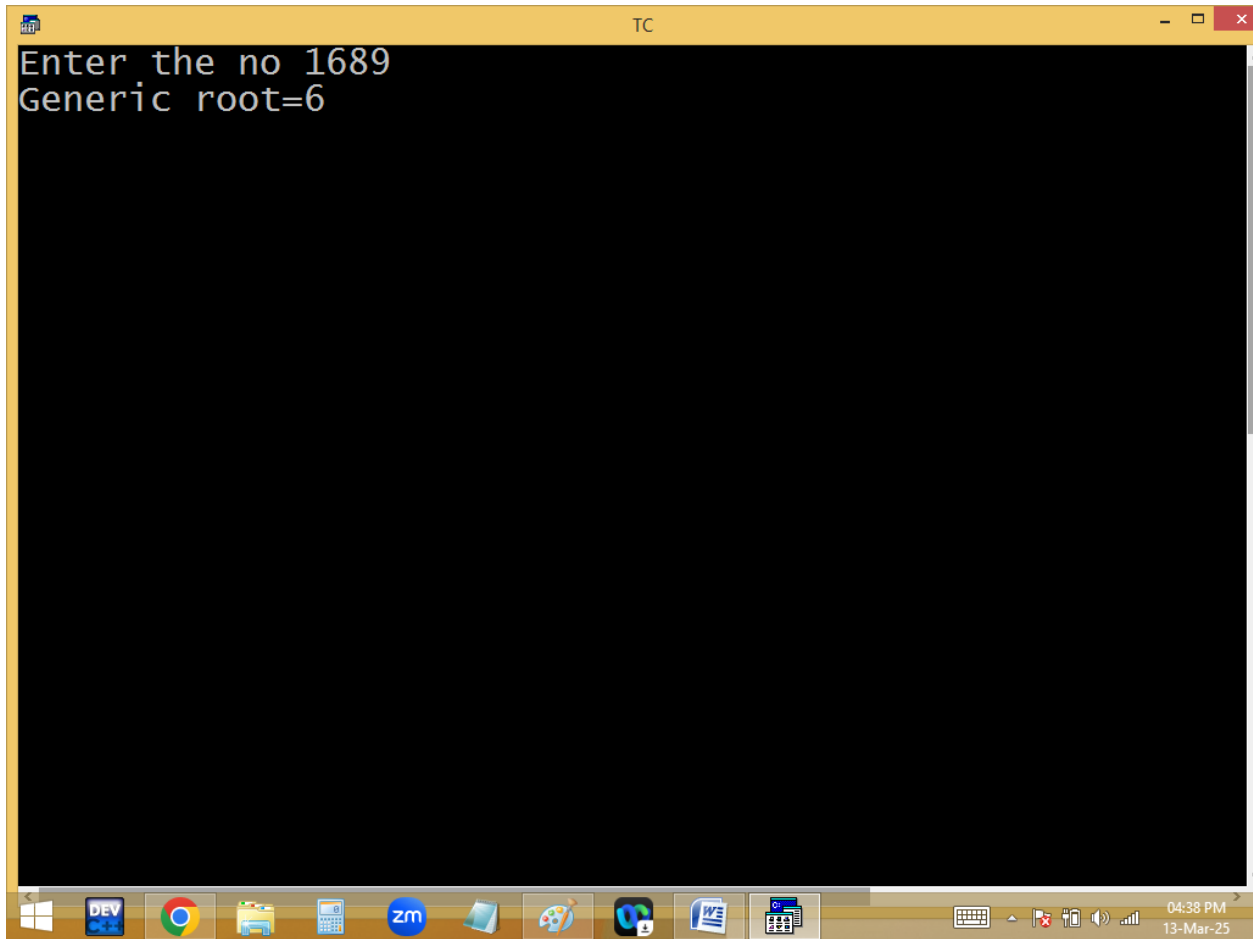


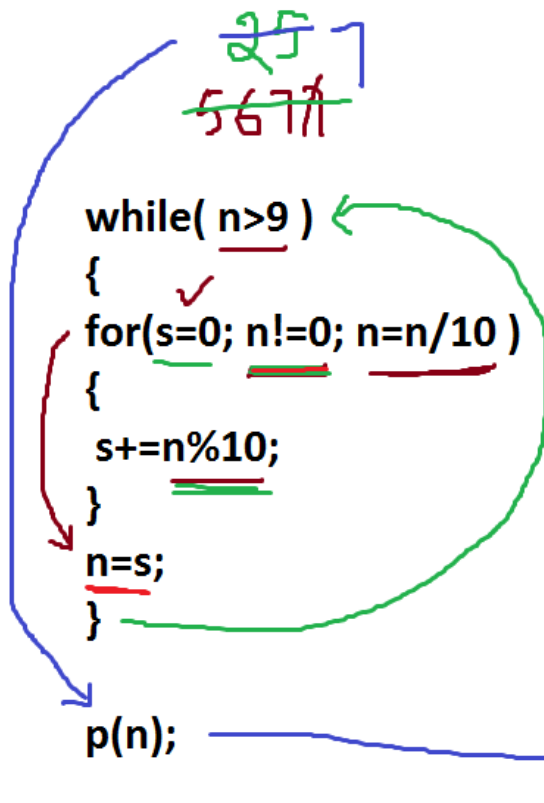
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 13", "Col 29", and lists editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. It contains a C program that calculates the generic root of a number. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n;int s;
    clrscr();
    printf("Enter the no "); scanf("%ld",&n);
    while(n>9)
    {
        for(s=0; n!=0; n=n/10) s+=n%10;
        n=s;
    }
    printf("Generic root=%d",n);_
    getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "04:38 PM" and "13-Mar-25".

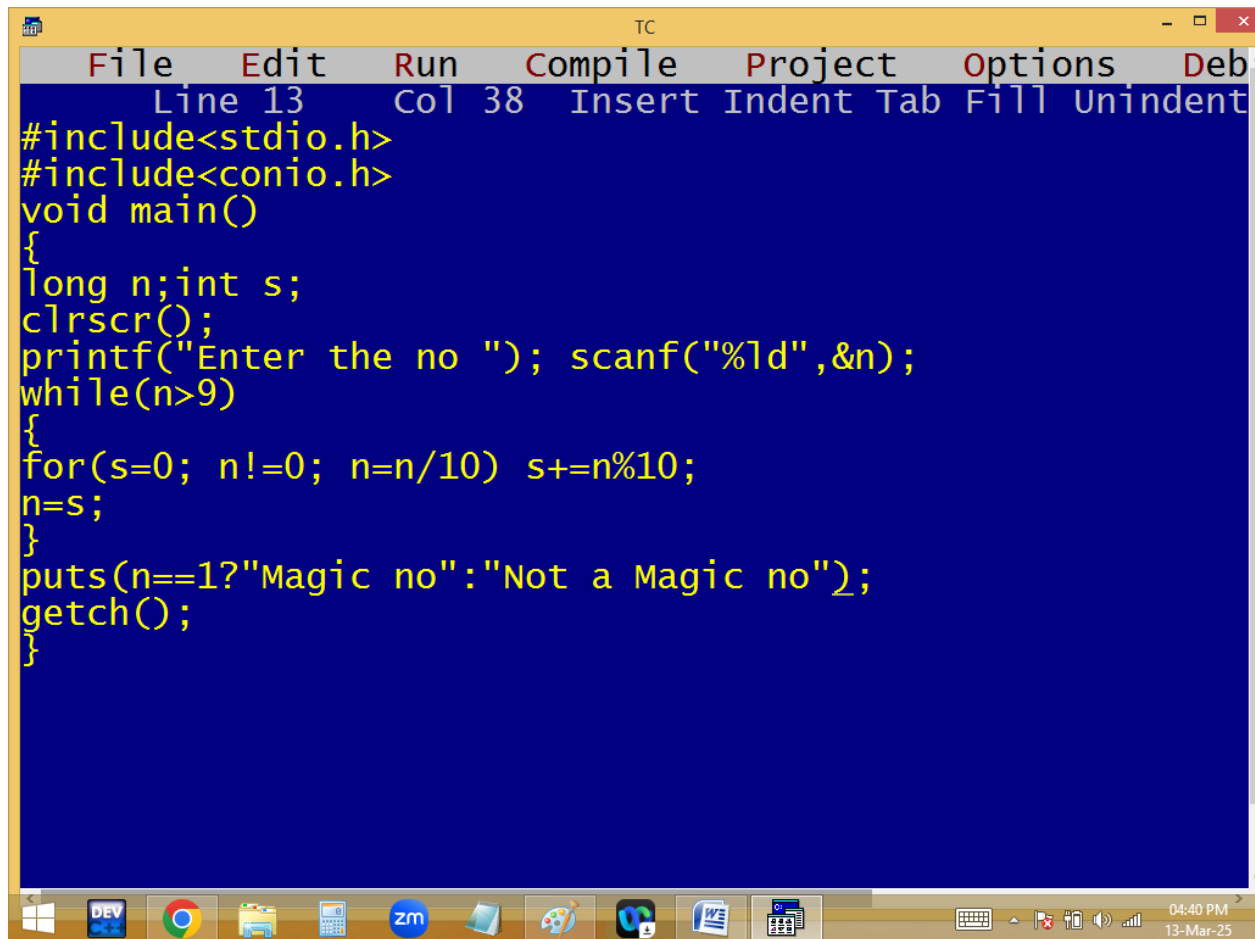






$$\begin{array}{rcl}
 \frac{n}{5677} & \frac{s}{0} & \\
 5677 \% 10 = 7 + 0 = 7 & & \\
 567 \% 10 = 7 + 7 = 14 & & \\
 56 \% 10 = 6 + 14 = 20 & & \\
 5 \% 10 = 5 + 20 = 25 & & \\
 25 \% 10 = 5 + 0 = 5 & & \\
 2 \% 10 = 2 + 5 = 7 & &
 \end{array}$$

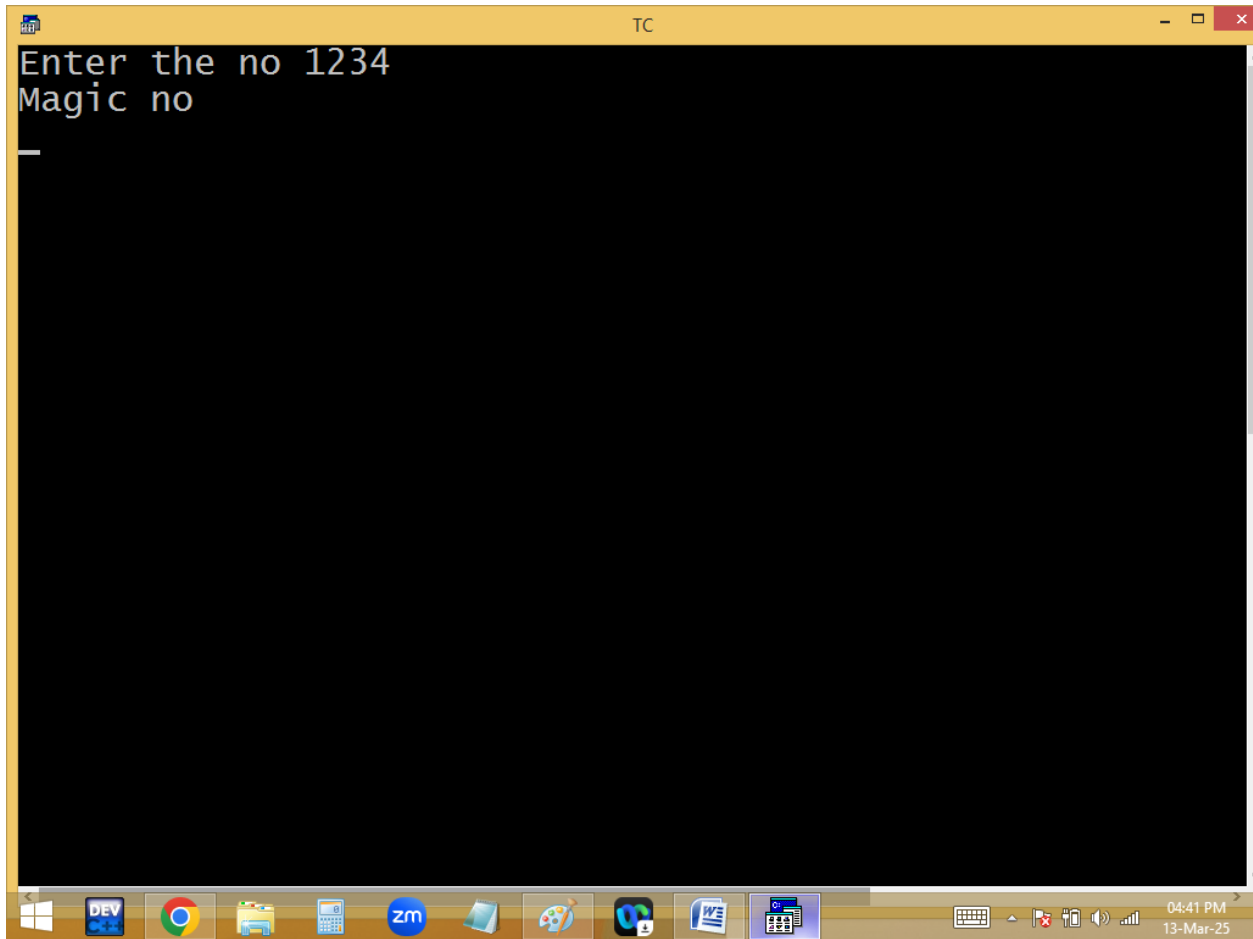
Finding magic no or not? If the generic root is 1 it is a magic no.

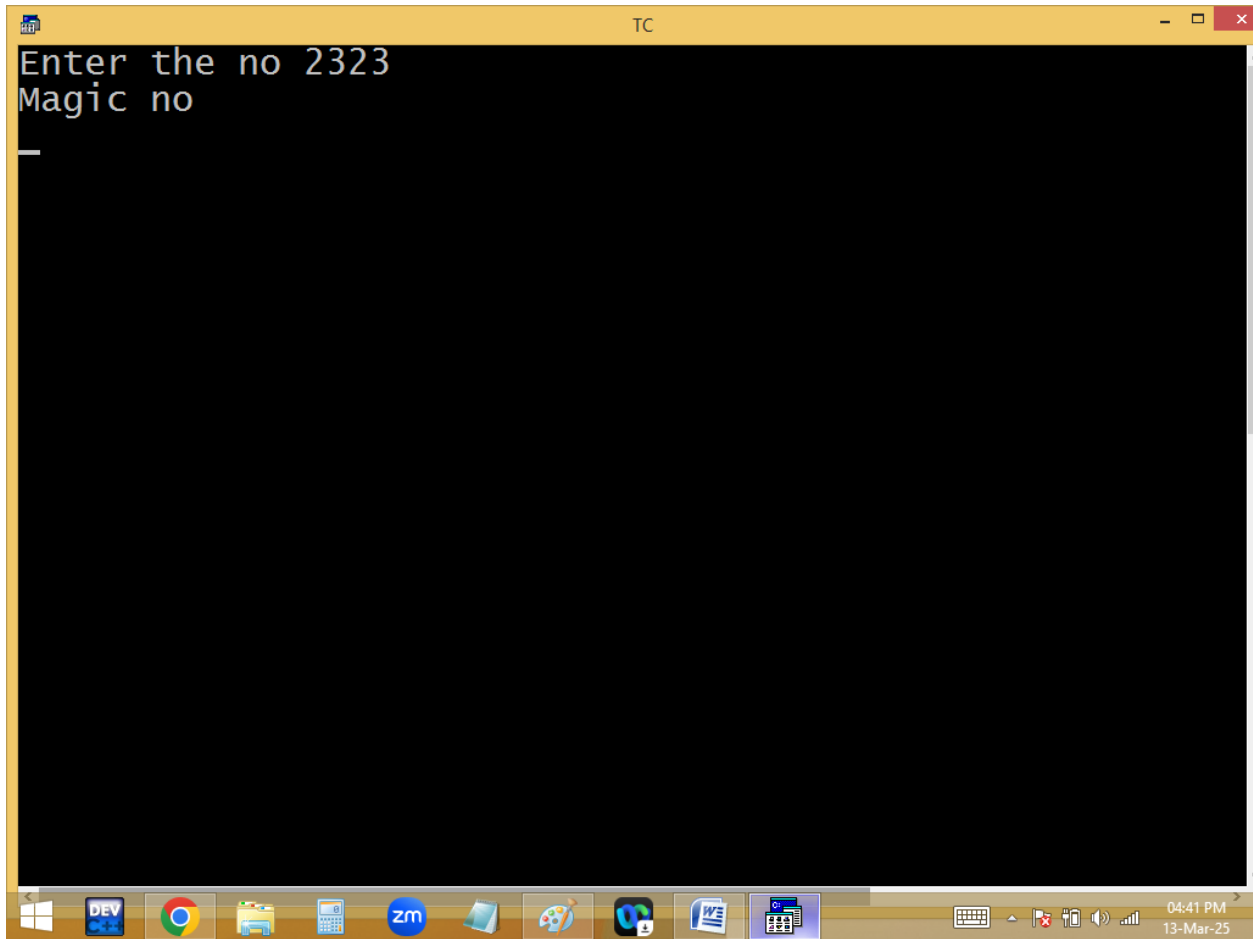


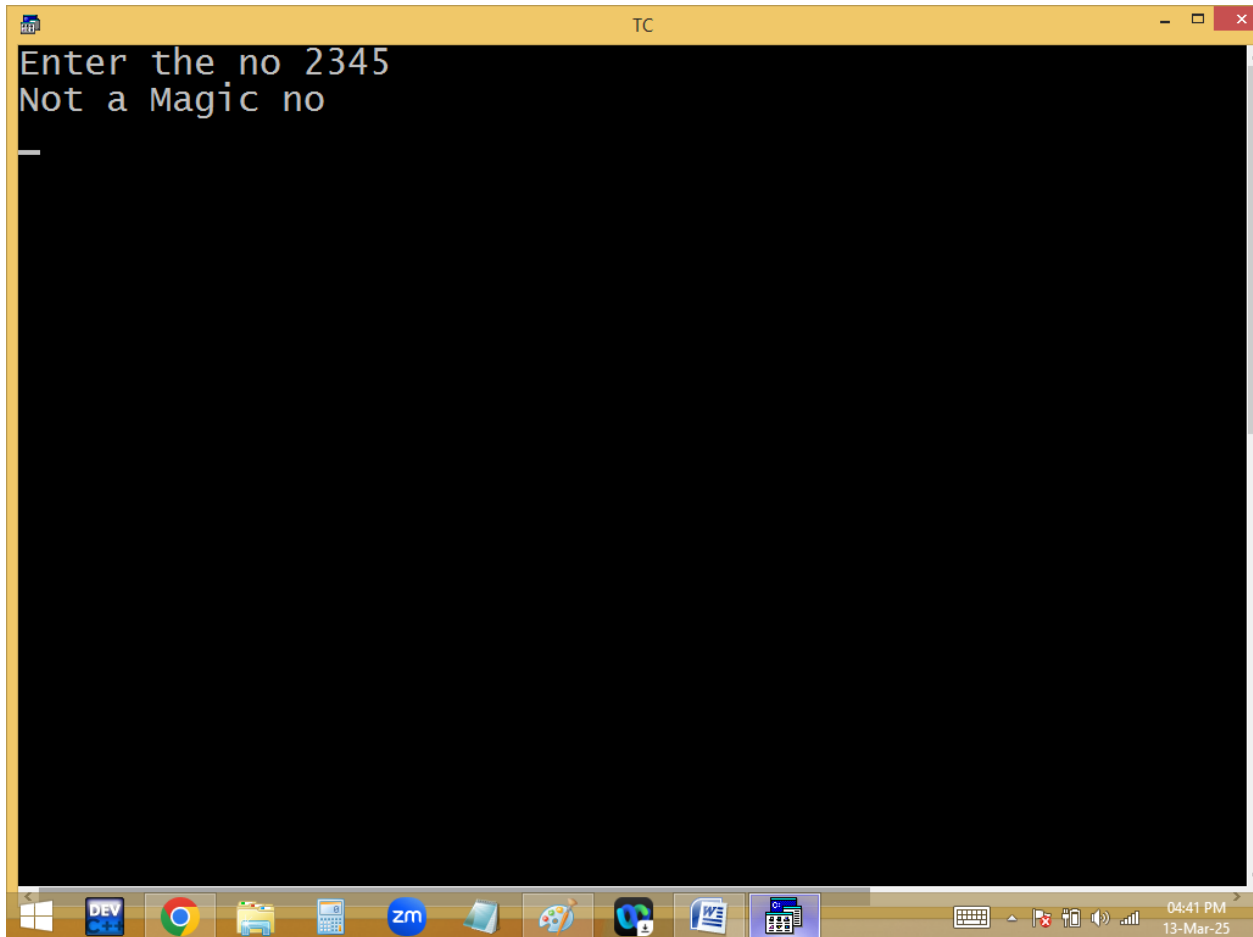
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 38", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. The code is a C program that checks if a number is a "magic number". It includes `<stdio.h>` and `<conio.h>`. The `main` function declares `long n` and `int s`, calls `clrscr()`, and prompts the user to "Enter the no ". It then enters a `while` loop that continues as long as `n > 9`. Inside the loop, it calculates the sum of the digits of `n` using a `for` loop: `for(s=0; n!=0; n=n/10) s+=n%10;`. After the loop, it assigns `n=s` and uses a ternary operator to print either "Magic no" or "Not a Magic no". Finally, it calls `getch()` to wait for a key press before exiting.

```
File Edit Run Compile Project Options Deb
Line 13 Col 38 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
long n;int s;
clrscr();
printf("Enter the no "); scanf("%ld",&n);
while(n>9)
{
for(s=0; n!=0; n=n/10) s+=n%10;
n=s;
}
puts(n==1?"Magic no":"Not a Magic no");
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner indicates the time is 04:40 PM on 13-Mar-25.







```
TC
Enter the no 2345
Not a Magic no
_
```

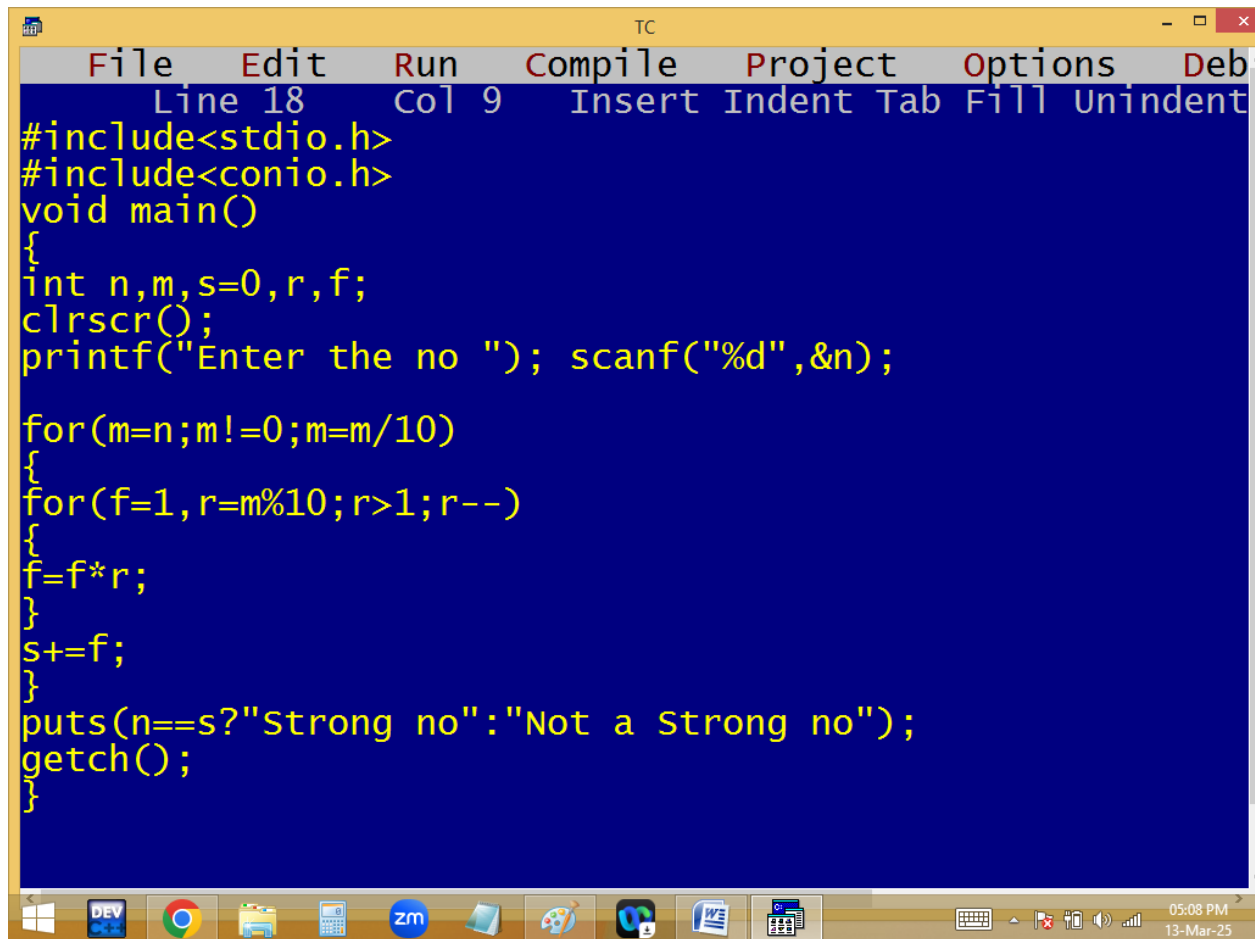
Finding strong no or not?

$$1 \rightarrow 1! = 1$$

$$2 \rightarrow 2! = 2$$

$$145 \rightarrow 1! + 4! + 5! = 1 + 24 + 120 = 145$$

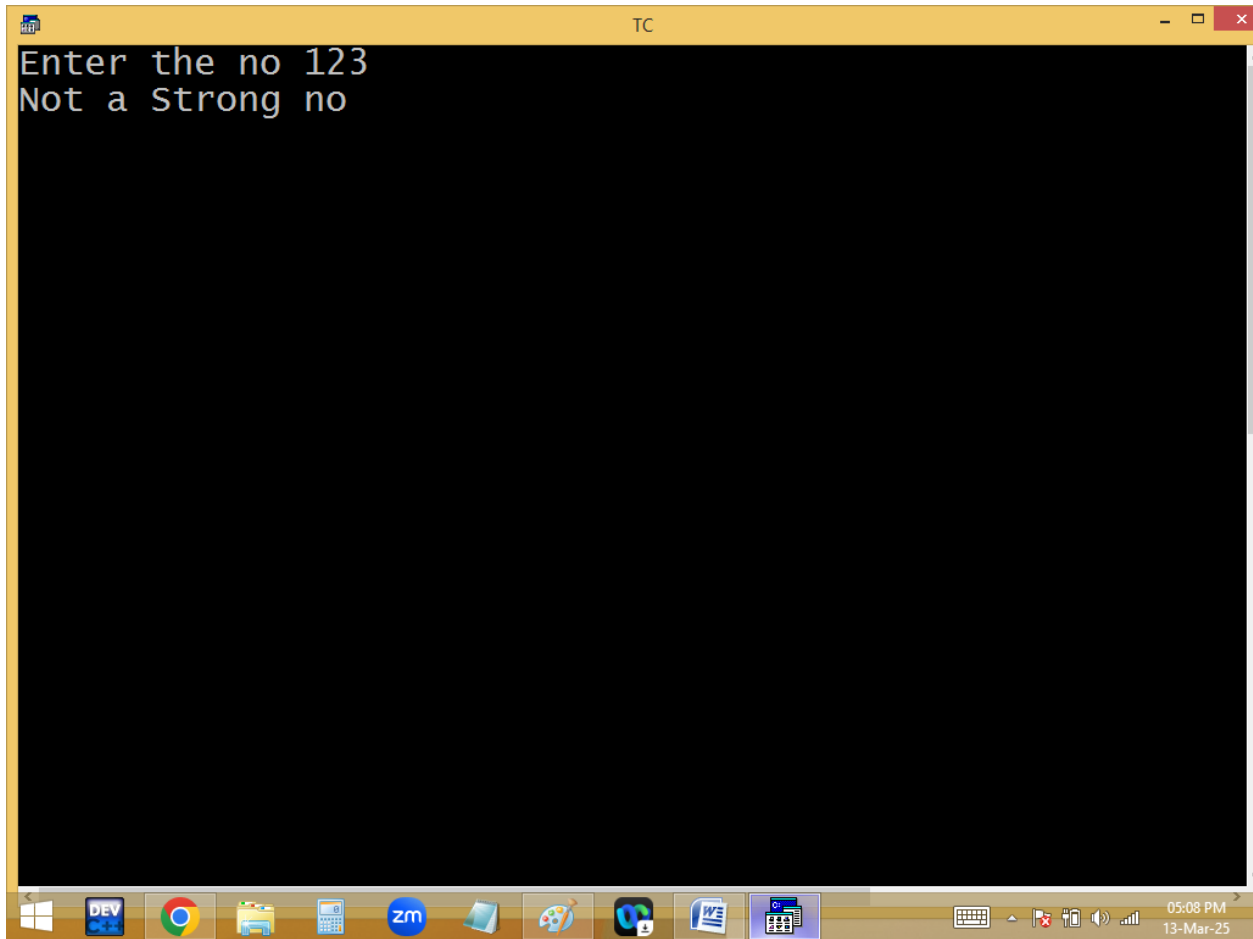
$$3 \rightarrow 3! = 6 \leftarrow \text{Not a strong no}$$

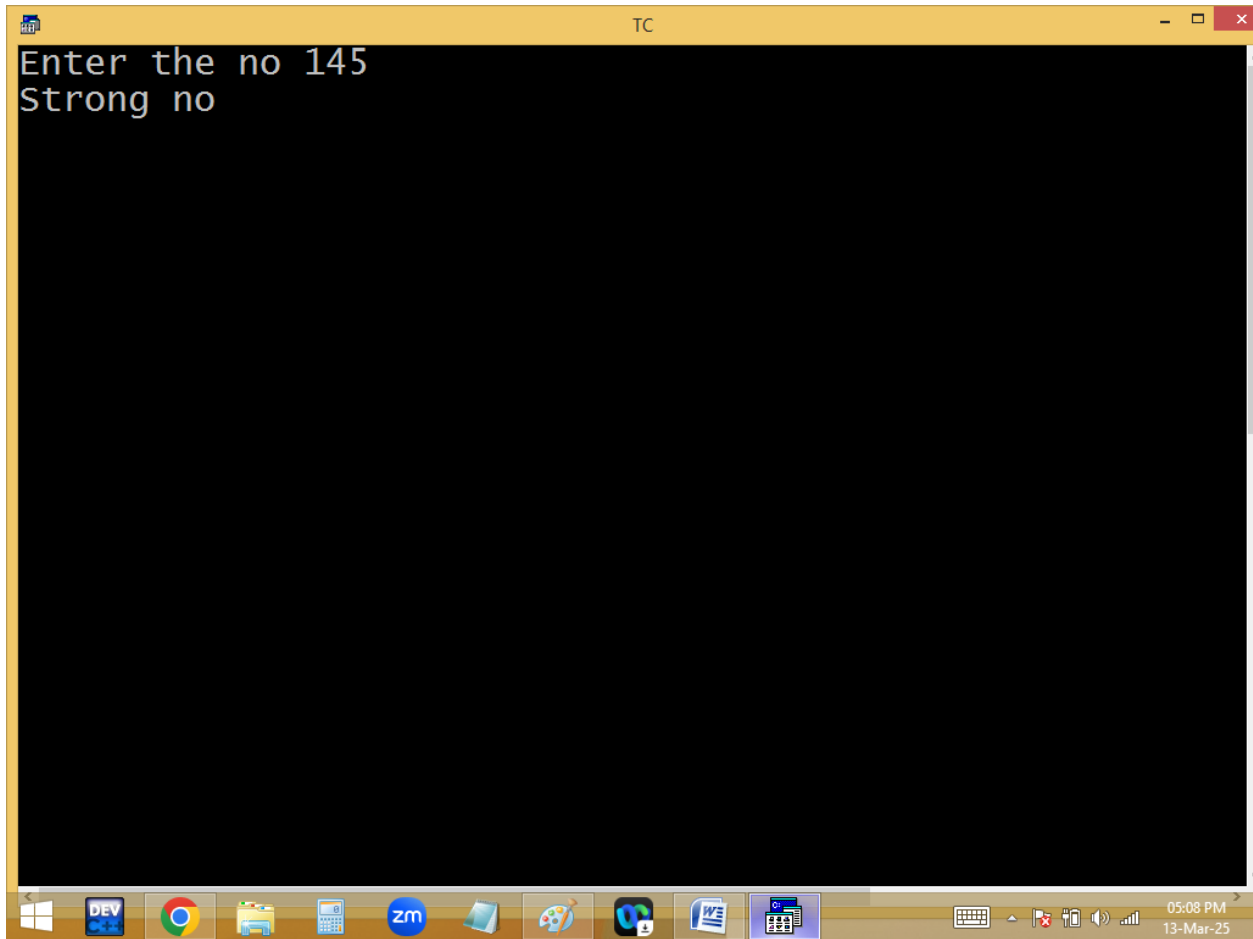


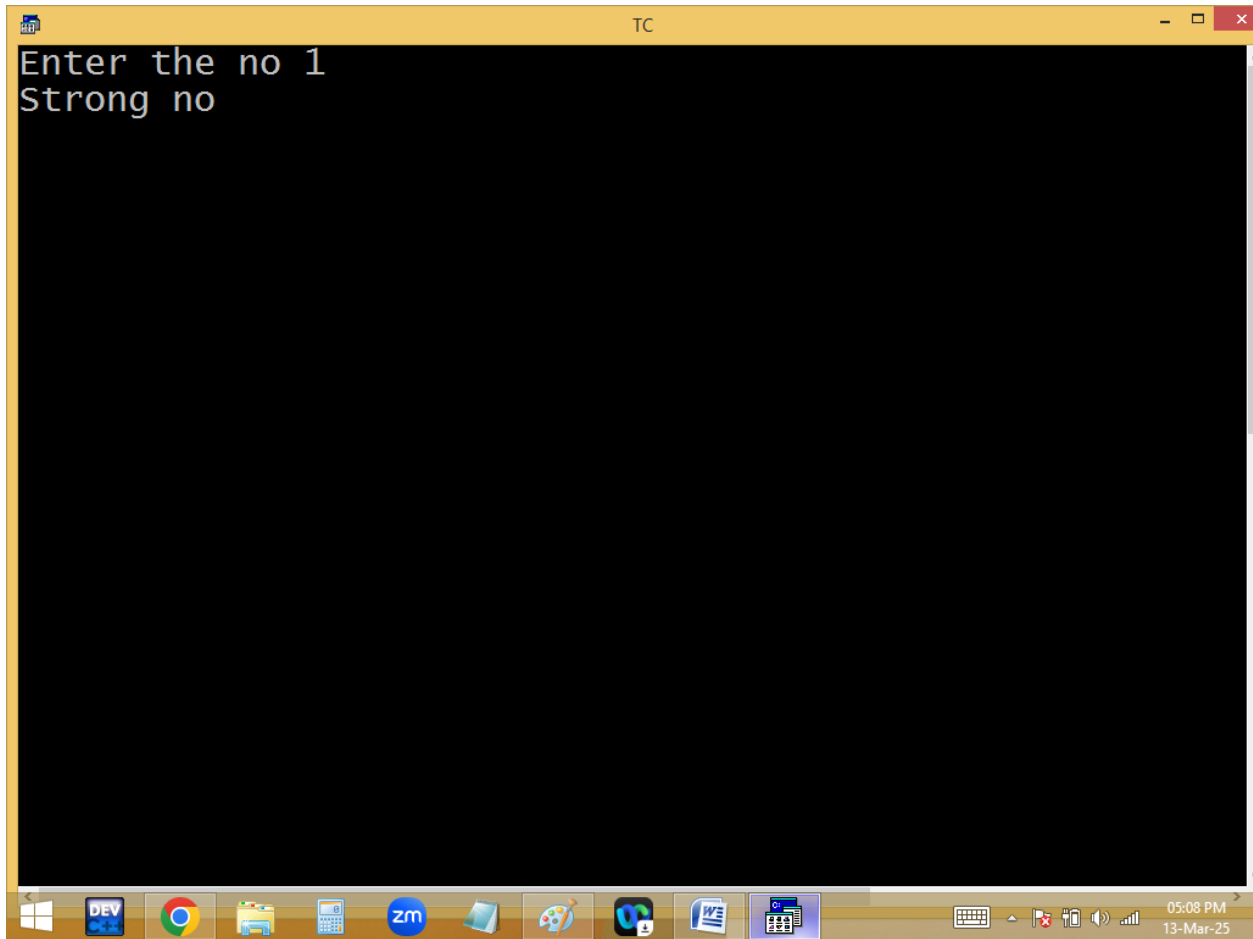
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 18", "Col 9", and lists editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. It contains a C program that checks if a number is a strong number. The code includes headers for `stdio.h` and `conio.h`, defines a `main` function, declares variables `n, m, s=0, r, f`, clears the screen with `clrscr()`, prompts the user to enter a number, and uses nested loops to calculate the sum of the factorials of its digits. It then compares the sum with the original number to determine if it is a strong number. The Windows taskbar at the bottom shows various icons including the Start button, DEV, Chrome, File Explorer, Calculator, Zoom, and others. The system clock in the bottom right corner shows "05:08 PM" and "13-Mar-25".

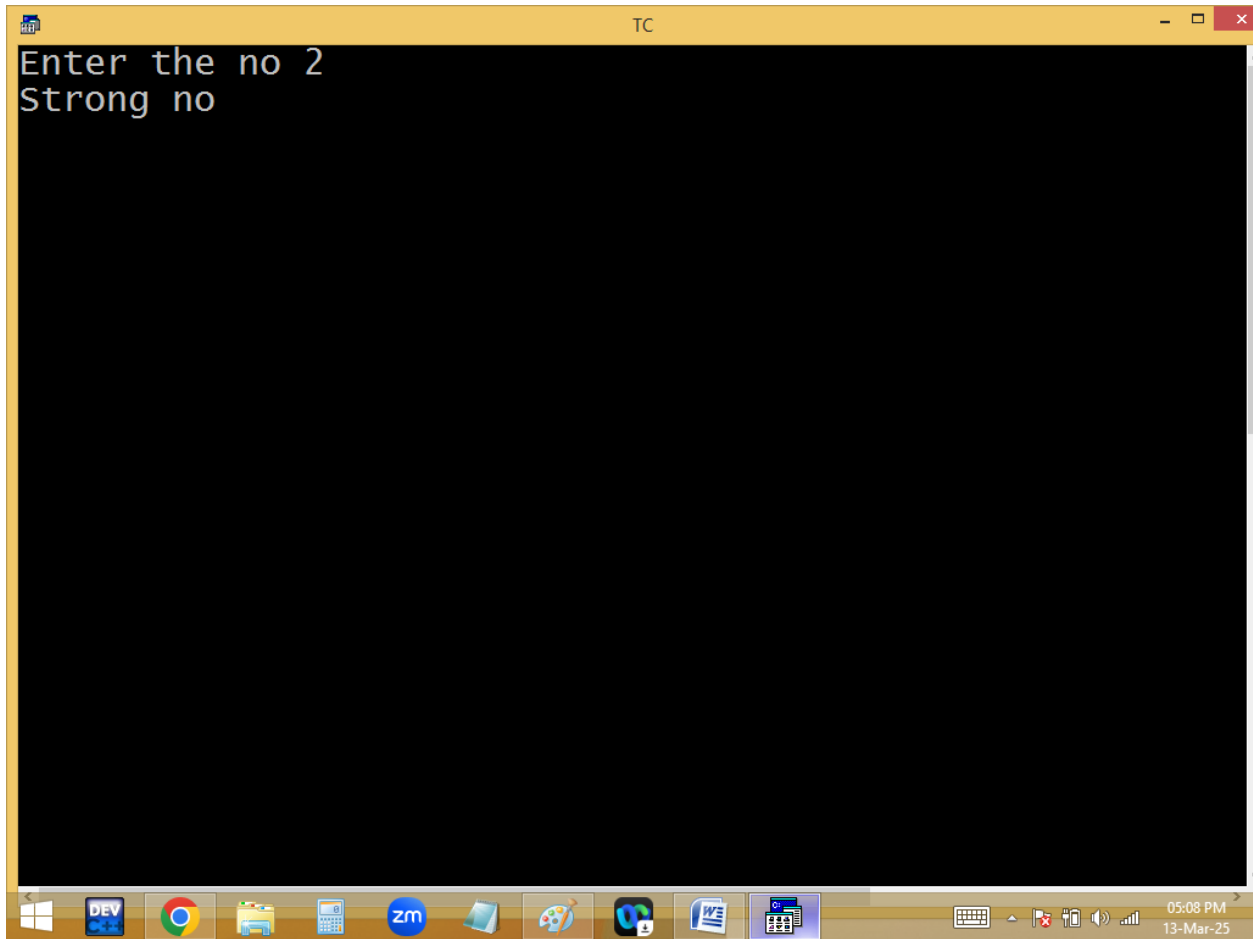
```
File Edit Run Compile Project Options Deb
Line 18 Col 9 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n,m,s=0,r,f;
clrscr();
printf("Enter the no "); scanf("%d",&n);

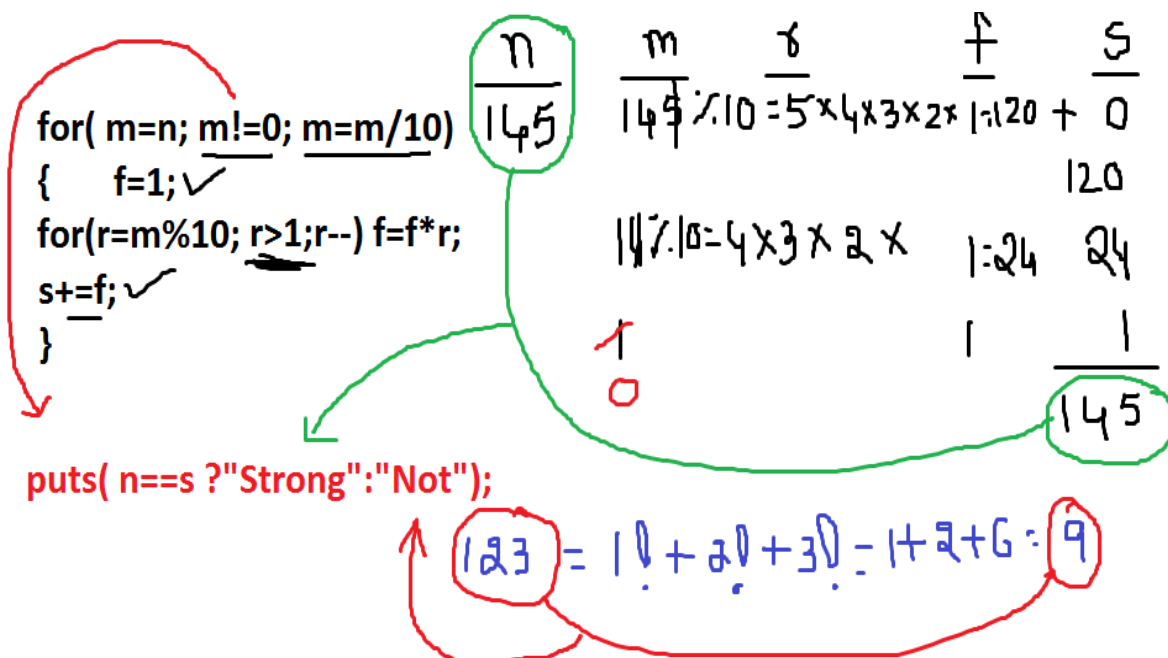
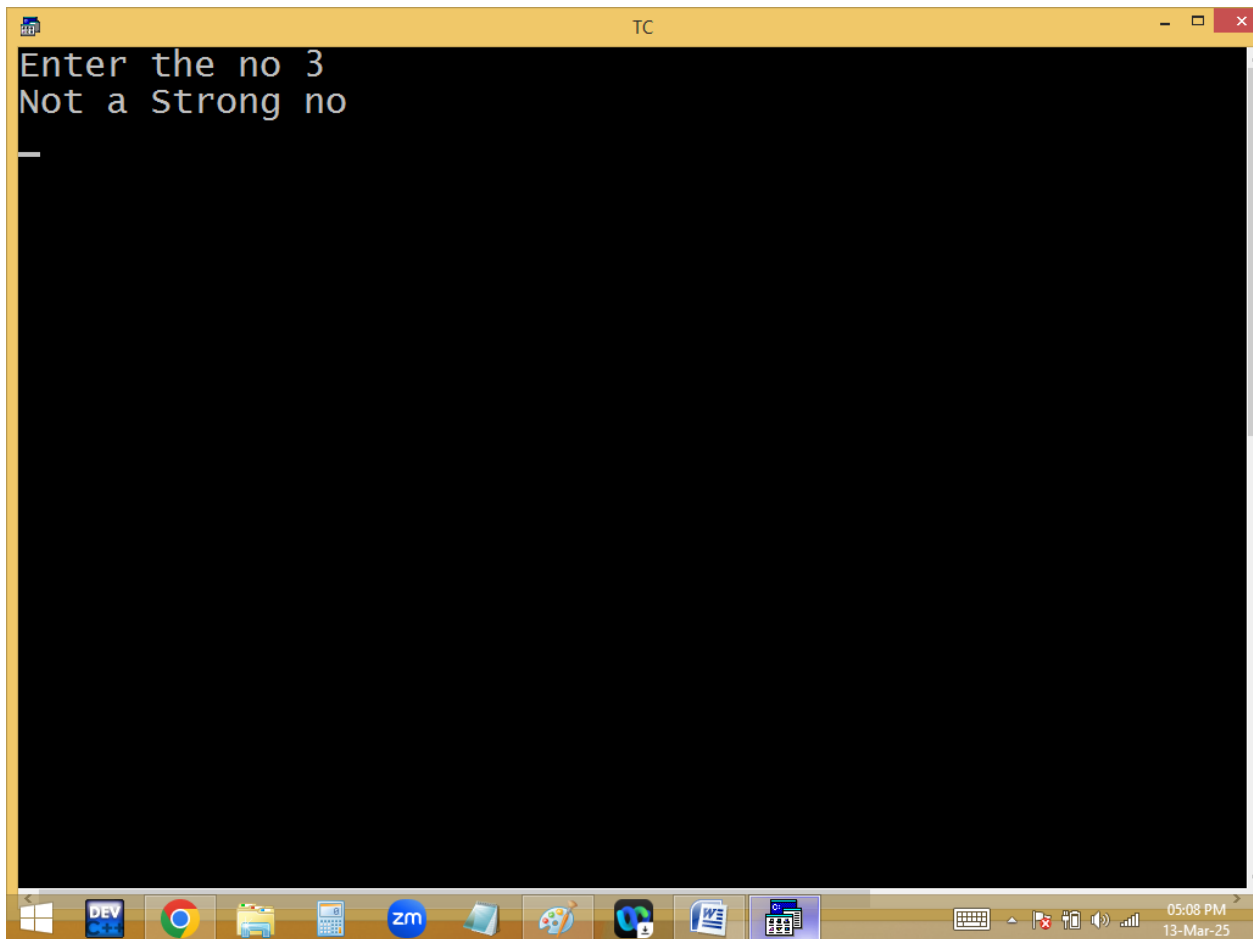
for(m=n;m!=0;m=m/10)
{
for(f=1,r=m%10;r>1;r--)
{
f=f*r;
}
s+=f;
}
puts(n==s?"Strong no":"Not a Strong no");
getch();
}
```



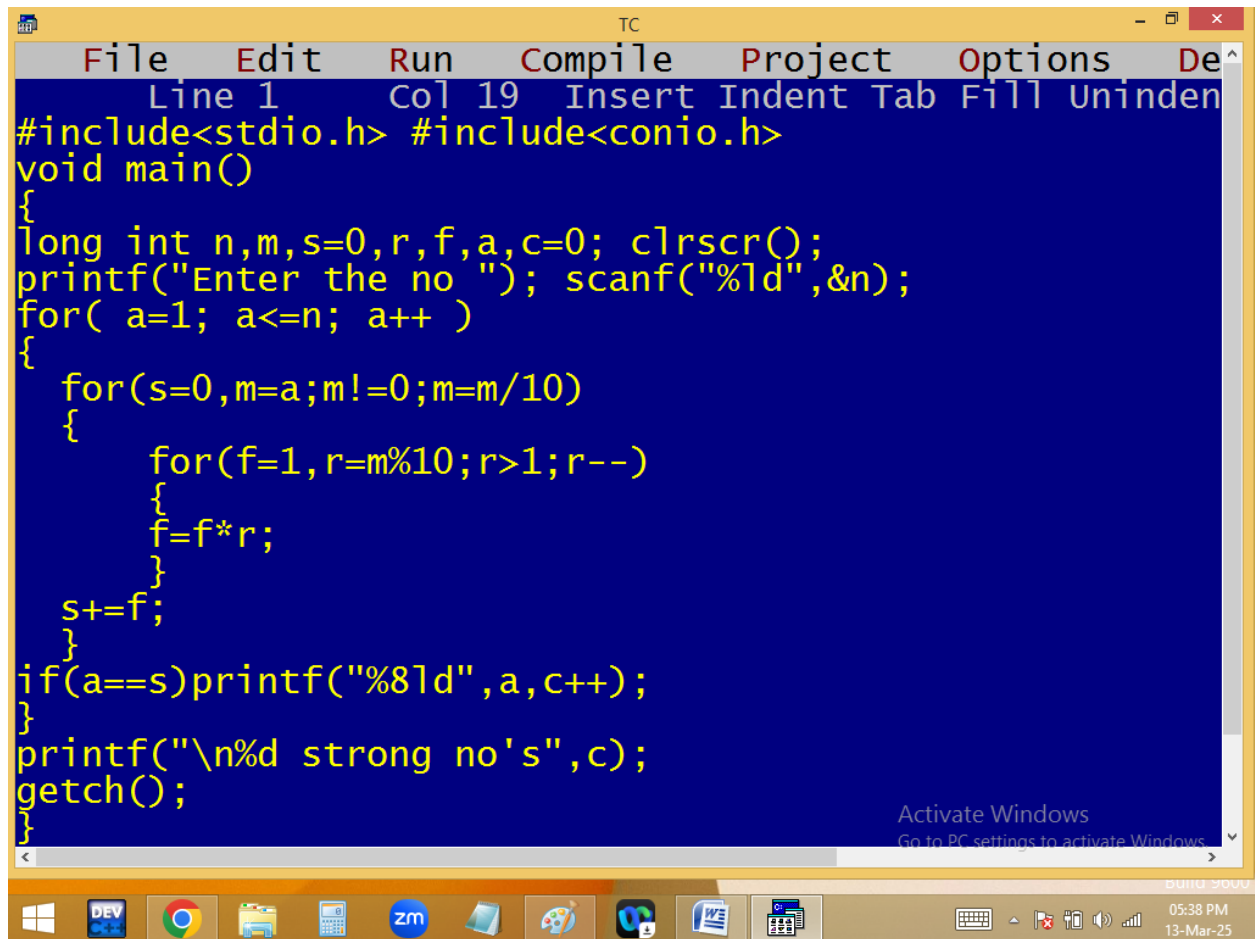








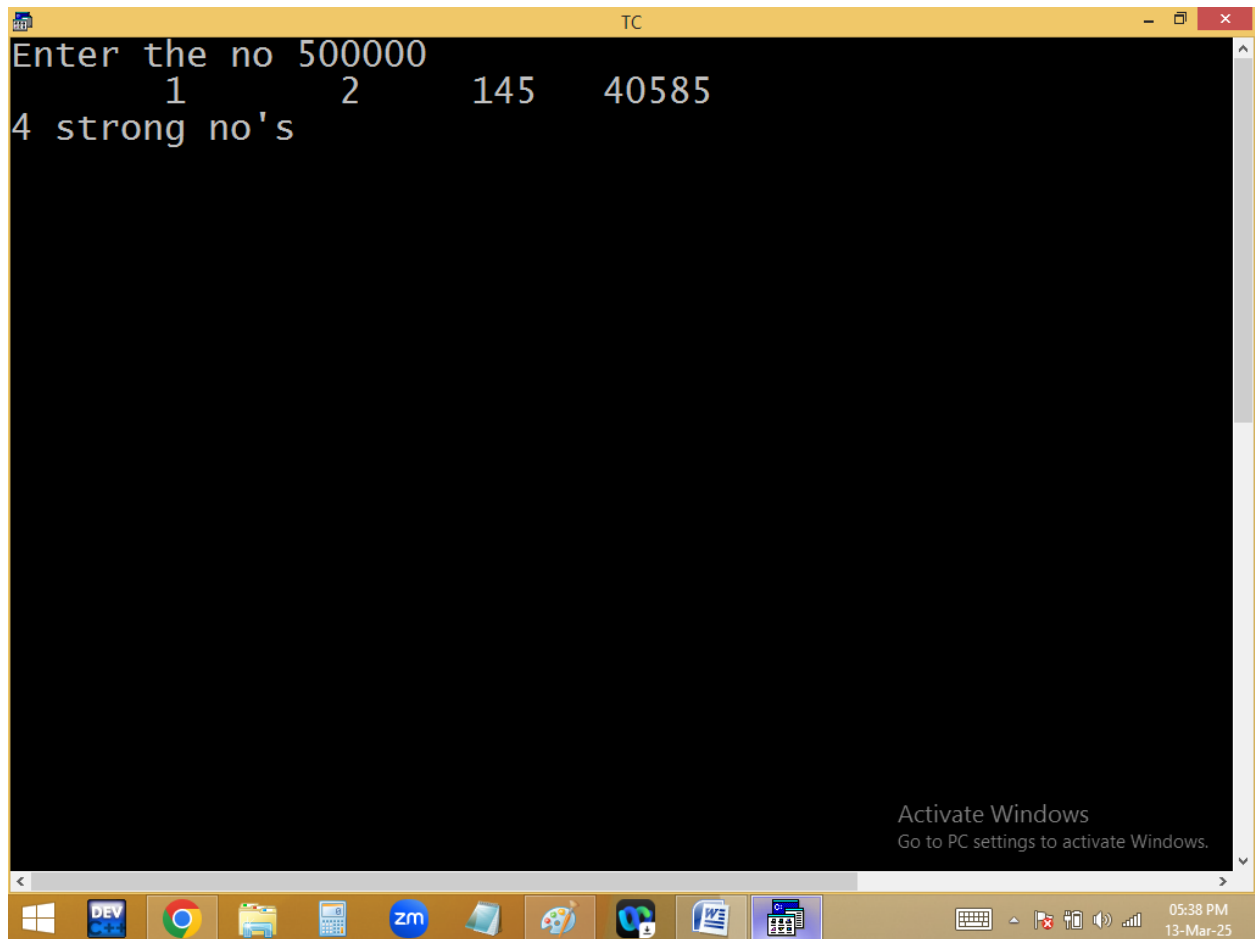
Printing 1..n all the strong no's



```
TC
File Edit Run Compile Project Options De
Line 1 Col 19 Insert Indent Tab Fill Uninden
#include<stdio.h> #include<conio.h>
void main()
{
long int n,m,s=0,r,f,a,c=0; clrscr();
printf("Enter the no "); scanf("%ld",&n);
for( a=1; a<=n; a++ )
{
    for(s=0,m=a;m!=0;m=m/10)
    {
        for(f=1,r=m%10;r>1;r--)
        {
            f=f*r;
        }
        s+=f;
    }
    if(a==s)printf("%8ld",a,c++);
}
printf("\n%d strong no's",c);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows

05:38 PM
13-Mar-25



```
Enter the no 500000
1      2      145      40585
4 strong no's
```

The screenshot shows a Turbo C++ (TC) window with a black background and white text. The window title bar says 'TC'. The program prompts the user to 'Enter the no' and receives the input '500000'. It then outputs the numbers '1', '2', '145', and '40585' on the same line, followed by a new line containing '4 strong no's'. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 05:38 PM on 13-Mar-25.

Home work:

1. Printing 1..n palindrome no's and count
2. Printing 1..n primes and count
3. Printing n..n primes and count
4. Finding twisted prime or not

13 is a prime and 13 reverse 31 also prime